

My passion for Computer Science originated from my belief that it makes the world more interconnected and assists the world by improving our technology and inventing new machines to solve environmental problems. I believe that pursuing a career in Computer Science would allow me to work to benefit society and improve our livelihoods and future.

My IB Physics Extended Essay was about thermodynamics, discovering how different heat transfer methods affected temperature loss through Fourier's Law and Stefan-Boltzmann's Law. I created simulations for the heat transfer methods to analyse the variables that affect the heat flow. In Mathematics, I understood more about the real-life applications of maths, specifically utilising probability to find out the different outcomes of an event. This enlightened me about the risks of every decision we make, benefitting me when making algorithms for predicting events such as how likely tomorrow would rain, while also training my problem-solving skills.

In Geography, I learnt about Eco-Cities and Smart Cities, which made me discover the varied uses of Computer Science, deepening my interest in this subject. I found out how governments utilise new technologies such as big data and 5G to improve the efficiency and effectiveness of recycling and improving the standard of living of cities.

My interest lies in STEM and especially in programming. The beauty of programming is the flexibility to create a similar outcome with different programming and methods. Learning Python, PowerBI and Tensorflow made me understand more about how people build software and also helped me in life. Doing this has caused me to understand more about the work of a computer scientist, and made me realise my interest in writing code. I wanted to learn how to code because it is a valuable and general skill to have in the future. Knowing code is integral in Computer Science, so by having this skill, I am more equipped to face any difficulties during the course,

Participating in the STEM team has also given me valuable experiences. Teaching primary school students about coding has improved my fundamentals in programming. Through working in the STEM team, I have become more proficient in communicating and working together with my clubmates, collaborating to build and create projects, troubleshooting bugs and mistakes, and discussing to improve our work. As Computer Science is a field where group work is integral for success, this skill would be beneficial when learning about it, improving the efficiency and quality of work.

Lastly, I have also learnt how to play the piano, handbell, and hand chimes, improving my coordination and teamwork. Playing badminton and archery has helped my focus and stamina when working. I also like to play video games and read, making me more observant and patient, improving my work and learning. I have also gone to various charity organisations such as Food Angel and BGCA to assist disadvantaged people, which has made me understand more about the problems we face, and how Computer Science may assist in solving these problems.

When I was a manager of the School Cafe to contribute to my school, I understood the importance of organisation and planning when working inside an establishment. I have acquired leadership and management skills, facilitating the daily workflow and contacting suppliers.

In conclusion, I believe that I am a suitable candidate for Computer Science at HKUST due to multiple reasons; expertise in coding, logical reasoning, and analytical skills, coming from rich experiences in different disciplines and interests, and I hope that you may accept my application.